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# PRACTICAL & TECHNICAL VIVA QUESTIONS

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## Basic Functional Questions

These test your understanding of what the app *does*.

1. What happens when a user clicks “Search”?
  2. How is the filter implemented on the search results page?
  3. How does the “Submit PG” form work — what happens after clicking submit?
  4. How do you fetch and display PG listings from Firestore?
  5. Can you describe how real-time chat works in your system?
  6. How do you prevent users from submitting blank or incorrect forms?
  7. How do you display only PGs that match the selected filters?
  8. What is the flow from logging in to sending a message?
  9. How does the app differentiate between a “user” and an “owner”?
  10. What will a new owner see after signing up and logging in?
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## Medium Technical Questions

These test logic, implementation, and architecture.

11. How did you structure Firestore collections and documents?
  12. How did you enforce role-based restrictions (both in frontend and Firestore rules)?
  13. How does `useEffect` help in your PG listing or chat components?
  14. How are images stored and retrieved securely using Firebase Storage?
  15. How is authentication handled on both the frontend and backend?
  16. What are the Firestore read/write rules you wrote?
  17. How do you ensure that only owners can edit their own listings?
  18. What are the edge cases you tested for in chat or room submission?
  19. How do you fetch dynamic route data (e.g., `/details/:id`) in React?
  20. How is the user redirected after login? How is session maintained?
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## Deep Technical/Advanced Questions

These test your architectural choices and understanding of scale, security, and performance.

21. How would you implement an admin panel to moderate listings?
  22. How do you plan to prevent spam messages in chat?
  23. What are the security concerns of using Firebase in production?
  24. How would you handle failed image uploads or internet loss during submit?
  25. How scalable is your current Firebase setup? What limits exist?
  26. How would you integrate Razorpay or Google Maps in your project?
  27. How would you handle user feedback and ratings without abuse?
  28. How does Firestore differ from a traditional SQL database in use-case?
  29. If a user bookmarks PGs, where and how do you store that data?
  30. What happens if a user reloads the page while inside a room details page?
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## React and Component Logic

These test your JavaScript + React knowledge.

31. What is the difference between controlled and uncontrolled components in React?
  32. How do you handle multiple inputs in a form using React?
  33. Why is `useEffect` called with dependencies?
  34. How is conditional rendering used in login, roles, or submit pages?
  35. How is React Router's `useParams` or `useNavigate` used in your app?
  36. How did you handle error messages and success feedback?
  37. What are the advantages of component-based architecture?
  38. What is `props drilling` and did you experience it?
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## Security & Access Control (Applied Security)

39. How do you secure PG listing edits so only the owner can change them?
  40. How do you secure Firestore collections from public access?
  41. Can users modify Firestore data from the browser dev tools?
  42. What steps would you take if someone reverse-engineered your frontend?
  43. How is chat data protected – can other users access it?
  44. How are passwords stored in Firebase?
  45. Is your login system vulnerable to brute force or phishing?
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## Testing, Debugging, & Maintenance

46. How did you test your system before final submission?
47. What tools or browser features did you use for debugging?
48. How would you identify and fix a bug if a PG listing fails to upload?

49. How do you test responsiveness and design consistency?
50. What tools or methods would you use for app performance monitoring?
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## Bonus Real-World Scenarios

51. What if Firebase raises its pricing or becomes unavailable?
52. How would you migrate data to MongoDB or another system?
53. Can this system be turned into a mobile app? How?
54. How would you handle multi-language/localization features?
55. Can you generate a report of all PG listings by owner ID?
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## Final Tips for Handling Questions

- **Think out loud:** If you don't know, explain your reasoning.
  - **Be honest:** It's okay to say "We didn't implement it, but here's how I would."
  - **Connect to your work:** Always relate answers to what **you actually built**.
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